

Anutam Srinivasan

+1-614-902-2763 | asrinivasan350@gatech.edu | anutamks.github.io

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Ph.D. in Robotics, President's Fellowship Awardee

Aug. 2025 – Present

Ohio State University

Columbus, OH

BS in Electrical & Computer Engineering; Mathematics (double major), GPA 3.993/4.000

Aug. 2021 – May 2025

EXPERIENCE

Trustworthy Robotics Lab, Georgia Tech

Aug. 2025 – Present

Research Assistant

Atlanta, GA

Robust Planning Out-of-Distribution with Learned Dynamics, *PI: Prof. G. Chou*

Aug. 2025 - Present

- Developed a fast approach for robustly planning with learned dynamics despite model errors.
- Used Conformal Prediction (CP), an uncertainty quantification method that produces confidence sets to provide probabilistic bounds on model error.
- Leveraged System Level Synthesis, a robust control synthesis framework, by modeling the dynamics as a system with bounded stochastic noise as provided by CP.
- Empirically validated the framework using car and quadcopter dynamics models.

Autonomy & Navigation Technology Center

May 2023 – Aug. 2025

ORISE Research Intern, Air Force Institute of Technology (AFIT)

Wright Patterson Air Force Base, OH

Geomagnetic Diurnal Variation Prediction, *PI: Prof. A. Nielsen*

May 2023 - May 2024

- Developed novel algorithms to predict diurnal variations in the geomagnetic field to improve the viability of Crustal Magnetic Anomaly Navigation – a complementary navigation method for GPS-contested environments.
- Explored use of relevant machine learning models (kNN, Neural Nets) for this purpose.
- Designed a portable ground-station (hardware) for magnetic field data collection to integrate with the above work.

Aircraft Magnetic Noise Prediction, *PI: Prof. A. Nielsen*

June 2024 – Aug. 2025

- Developing a Physics-Informed time-series Machine Learning model that can predict the magnetic noise produced from an aircraft so it can be removed from sensor readings.
- Converted ~4400 raw avionics data features to a machine-learning friendly format.
- Leveraging Tolles-Lawson physics model as a basis for the model architecture.

CoSyNe Lab, Ohio State

Aug. 2023 – Aug. 2025

Research Assistant

Columbus, OH

Stealthy Data (audio) Exfiltration on Powerlines via Phone Charging Leakage, *PI: Prof. K. Athreya*

- Designed methods to identify power leakage patterns when a charging phone is actively used.
- Extracted and reconstructed relevant multi-modal signals to demonstrate plausible attack scenarios.
- Building the evaluation environment to evaluate various attack scenarios and compare them to baselines.

Data Mining Research Lab, Ohio State

Nov. 2023 – Aug. 2025

Research Assistant

Columbus, OH

Fairness Under Uncertainty for Graph Models, *PI: Dr. P. Maneriker & Dr. S. Parthasarathy*

- Utilizing Conformal Prediction (CP) – an uncertainty quantification method that produces confidence sets
- Proposed a key algorithm to directly leverage the CP framework and construct *fair* uncertainty sets.
- Demonstrated and explored the generality of the *fair* CP framework.
- Implemented various CP algorithms for a benchmarking study on Graph Conformal Prediction.

TEACHING

Fundamentals of Engineering Honors Program

Aug. 2022 – May 2025

Undergraduate Teaching Assistant

Columbus, OH

ENGR 1281.01H - Fundamentals of Engineering Honors: Intro to Matlab & C

Fall 2022, 2024, 2025

- Laboratory TA (2022): Assisted students in the lab portion of the course, and graded weekly lab reports.
- Office Hours TA (2024, 2025): I helped students with their lab reports and graded assignments.

ENGR 1282.02H - Fundamentals of Engineering Honors: Robot Project

Spring 2023, 2025

- Class TA (2023): Helped students work through assignments and taught them programming, design, and communication strategies for their robot project, and graded class assignments.
- Office Hours TA (2025): I helped students develop a robot to complete a sequence of tasks.

Intro to Digital Signal Processing

Aug. 2025 – Present

Graduate Teaching Assistant

Atlanta, GA

- Lab and Office Hours TA: I run the student digital signal processing labs, host office hours, grade homework, and proctor exams.

PROJECTS

Fundamentals of Engineering Honors Robot Project

Spring 2022

- Designed a robot with a four-person team to autonomously complete a specified course.
- Utilized CAD and C++ to create our robot and implemented a PID controller to improve our performance.
- Received a design award for our robust construction.

Real Time Robotics Systems Final Project

Spring 2024

- Considered Multi-Agent path planning algorithms for constrained settings (E.g. warehouses).
- Performed literature review to identify and adapt existing path planning algorithms for our setting.
- Ran experiments, to identify pros and cons of methods (including tree and graph-based methods).

X-Ray Communication Capstone Project, Advised by Dr. C. Ball, ElectroScience Lab

Fall 2024 - Spring 2025

- Building a hardware-software testbed for X-Ray communications.
- Exploring techniques in visual communications systems to build a fast and reliable testbed.
- Learning about various communication and encoding protocols to minimize bit-error rate.

PUBLICATIONS AND PRESENTATIONS

[1] **Conference Paper.** A. Vadlamani*, A. Srinivasan*, P. Maneriker, A. Payani, S. Parthasarathy. “A Generic Framework for Conformal Fairness”, *Thirteenth International Conference on Learning Representations* (2025). (*Equal Contribution).

[2] **Journal Paper.** P. Maneriker*, A. Vadlamani*, A. Srinivasan, Y. He, A. Payani, S. Parthasarathy. “Conformal Prediction: A Theoretical Note and Benchmarking Transductive Node Classification in Graphs”, *Transactions in Machine Learning Research* (2025).

[3] **Preprint.** A. Srinivasan*, A. Vadlamani*, A. Meghraz, S. Parthasarathy. “FedCF: Fair Federated Conformal Prediction”, arXiv preprint arXiv:2509.22907 (2025). (under review, *Equal Contribution)

[4] **Preprint.** A. Srinivasan, A. Nielsen. “Extending Temporal Disturbance Estimations For Magnetic Anomaly Navigation and Mapping” arXiv preprint arXiv:2510.15113 (2025). (under review)

[5] **Preprint.** A. Srinivasan, G. Chou. “Safety Beyond the Training Data: Robust Out-of-Distribution MPC via Conformalized System Level Synthesis” (2025). (under review, submitted to L4DC 2026)

[6] **Conference Presentation.** A. Srinivasan, A. Nielsen. “Temporal Anomaly Corrections for Magnetic Anomaly Navigation”, Presented at Joint Navigational Conference, Kentucky, (2024).

[7] **Conference Presentation.** A. Nielsen, L. Bergeron, A. Srinivasan. “Extended Geomagnetic Ground Reference Station Model and Noise Characterization”, Presented as a poster at the American Geophysical Union Fall Conference, California, (2023).

INVITED TALKS

- | | |
|---|-----------|
| Temporal Anomaly Corrections For Magnetic Anomaly Navigation | June 2024 |
| <ul style="list-style-type: none">Invited to present our work on Magnetic Anomaly Navigation to a leading MagNav group at the National Oceanic and Atmospheric Administration (NOAA). | |

AWARDS

- | | |
|---|-----------------------|
| President's Fellowship, Georgia Tech | Aug 2025 – Present |
| Summa Cum Laude, > 3.9 GPA | May 2025 |
| Dean's List (all undergraduate semesters) | Aug. 2021 – May 2025 |
| Maximus Scholarship (Merit-Based Scholarship) | Aug. 2021 – May 2025 |
| Scholarship Award for Robot Design, by Aptiv | Aug. 2022 – Dec. 2022 |

EXTRACURRICULARS

- | | |
|--|-----------------------|
| Ohio State University Tennis Club | Aug. 2021 – May. 2023 |
| Code 4 Community (Project Lead: Aug 2023 – May 2024) | Aug. 2021 – May. 2024 |
| Ohio State University Squash Club (Treasurer: May 2024 – May 2025) | Aug. 2023 – May 2025 |

COURSES & SKILLS

Relevant Coursework: Real-time Robotics, Intro to Machine Learning, Mathematical Stats and Probability, Computer Architecture, Intro to Real Analysis (1 and 2), Abstract Algebra (1 and 2), Partial Differential Equations, Math Methods in Relativity Theory (Applied Differential Geometry), Intro to Wireless Communications, Linear Systems & Controls*, Intro to Artificial Intelligence*. *=In Progress

Skills: Python (Scikit Learn, PyTorch), MATLAB, Java, C, C++, Git/GitLab